**Politecnico di Milano**

# Dipartimento di Elettronica, Informazione e Bioingegneria

**prof. Fabrizio Ferrandi**

## Advanced Algorithms and Parallel Programming I part–September 5th, 2024

**Polimi ID** _____ _____

**Surname** _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

**Exercise 1 (4 points)** _____

**Exercise 2 (4 points)** _____

**Exercise 3 (4 points)** _____

**Exercise 4 (4 points)** _____

**Exercise n. 1**

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. In a PRAM, $pE_p(n)$ is equal to the application speed-up? (1)

B. What happens to the PRAM model if we have a bounded number of processors smaller than the one required by the theoretical model? What happens to the speed-up if the application is a do-all kind of application and we have a bounded number of processors? (2)

C. Could you please discuss the following sentence? "Amdahl's law presupposes that the computing requirements will stay the same, given increased processing power. In other words, an analysis of the same data will take less time given more computing power". Is it true or false? What does it mean? (1)

**Exercise n. 2**

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the types of problems that the master theorem can solve and the assumptions under which the theorem holds. Apply the theorem to the matrix multiplication algorithm and explain why Strassen's version is more efficient. (2)

B. Please describe the partitioning algorithm used by the quick sort and its key properties. How is it randomized? (1) **This question can be skipped in case of a passed challenge.**

C. To which class does the Longest Common Subsequence Algorithm belong? How is its complexity evaluated? (1)

**Exercise n. 3**

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Why does randomized Treap structure have a complexity on average similar to a balanced binary tree? Treap required all priorities to be pairwise distinct. How can this property be satisfied? (2)

B. Please discuss how a Split is implemented in a Treap Data-Structure. What happen in case the k element is in the initial Treap? (2)

**Exercise n. 4**

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the competitive analysis. In which class of algorithm is used? Amortized analysis could be of some use; why? (2)

B. What is the Computed Table, and what is it used for in ROBDDs? Does it also have to do with the canonicity of ROBDD? (2)